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AGE AND DURATION OF THE RÉUNION GEOMAGNETIC POLARITY EVENT

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Palaeomagnetic studies on oriented samples from two sequences of olivine basalt lava flows on the island of Réunion together record the Réunion normal polarity event within the Matuyama reversed polarity epoch. Detailed K–Ar dating of the lavas indicates that the Réunion event has a mean age of 2.02±0.02 my with a duration estimated to lie within the range 10,000 to 50,000 yr. This could explain the rarity of detection of the event in deep sea sedimentary cores.

1. Introduction

Although the durations of the Brunhes, Matuyama and Gauss polarity epochs are relatively well known with boundaries at t=0.69, 2.41 and 3.32 my [1, 2], the age limits of some of the shorter duration polarity events are poorly defined. This is partly because of the limited precision of the K–Ar dating method, which when combined with conventional palaeomagnetic study of volcanic rocks, has enabled compilation of the polarity time scale [1, 3]. This experimental limitation is particularly important if polarity events are inferred by the combination of data from stratigraphically unrelated outcrops. To date, only the younger boundary of the Jaramillo event has been proposed by the dating of igneous rocks of opposite polarity in known stratigraphic sequence [4]. The boundaries of all other events have been inferred, and although data from deep sea sediments have provided valuable confirmation in some cases of the existence and relative durations of the events [5, 6], such results from deep sea cores are far too inconsistent and circularly involved with the independently established polarity time scale, to provide fine definition of the scale [7].

In this communication we present data from two sequences of basalts exposed in the walls of the canyon of the Rivière St. Denis on the island of Réunion, southwestern Indian Ocean (fig. 1), which have recorded a period of normal polarity with reversed polarity lavas above and below.

2. Previous work

The magnetic polarity history as recorded in the lavas of Réunion has unusual significance. As summarized by Watkins [7] a normal polarity event in the early Matuyama reversed polarity epoch (t = 0.69 to 2.41 my), was identified as the major event in the Matuyama mainly because of its frequent identification in deep sea sedimentary cores [8, 9]. This major event initially was correlated with the Olduvai normal polarity event, at about t = 1.9 my [10], which was also recognized in lavas of Alaska [11], and the Cocos Islands, Pacific Ocean [12]. Chamalaun and McDougall [13] and McDougall and Chamalaun [3] presented evidence from Réunion for a polarity change from normal (older) to reversed (younger) at t = 2.03 ± 0.05 my. The normal polarity lavas were tentatively correlated with the Olduvai event [10], although it was noted [3] that the Réunion ages were significantly older than the estimated age of 1.92 my assigned [10] to the Olduvai normal polarity basalts.

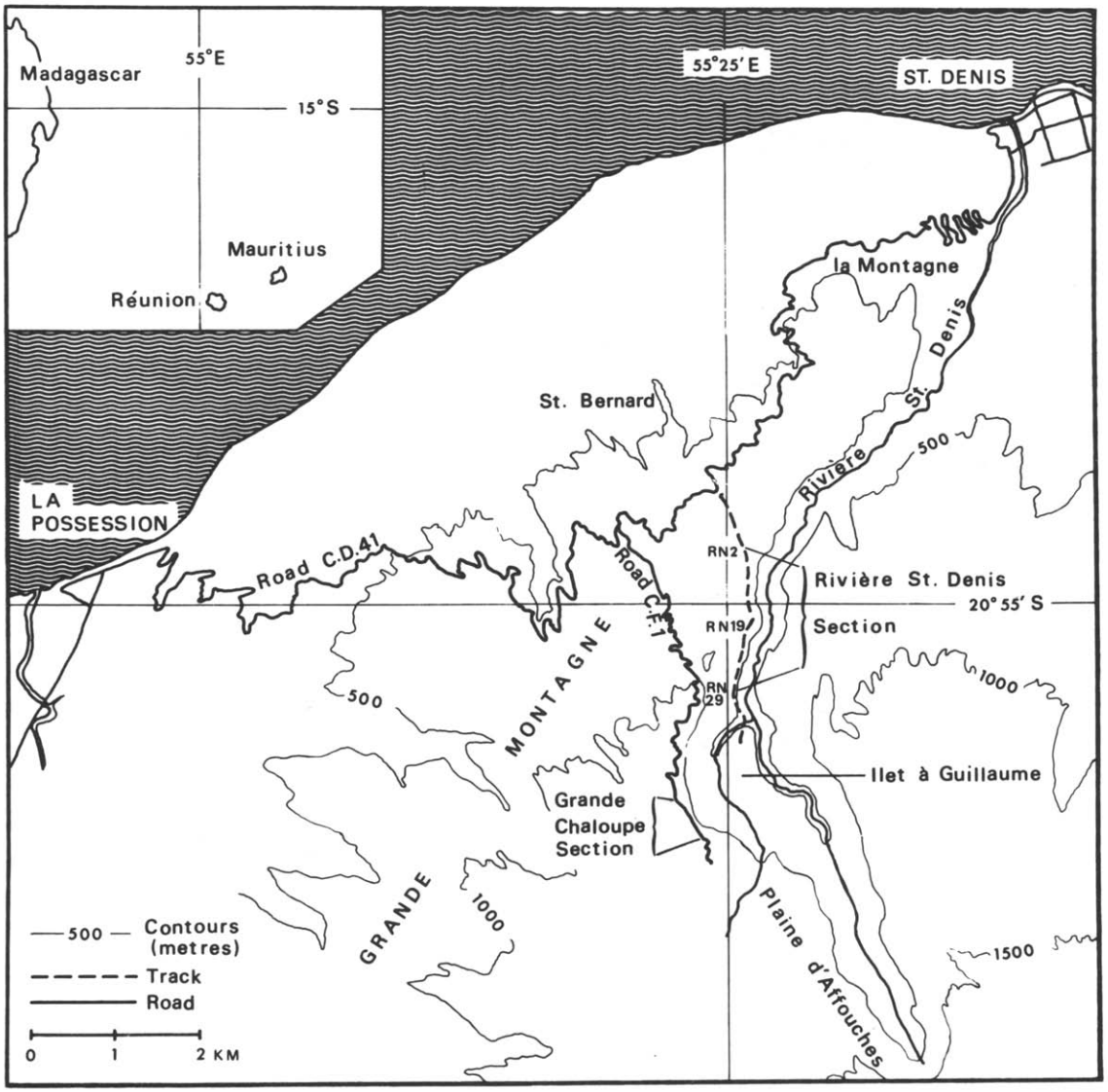


Fig. 1. Sketch map of the northwestern sector of Réunion indicating the location of the two sequences sampled for this study. Inset shows location of Réunion in the southwest Indian Ocean.

Later the major event in the early Matuyama was suspected to represent the Gilsá event, dated first on Icelandic lavas [14] at $t = 1.6$ my, and subsequently recognized in Alaska [15], Madeira [16] and Tenerife [17]. More recent studies by Grommé and Hay [18] show that the Olduvai event is younger than initially suspected with a mean age of about 1.72 my, and is possibly synchronous with the Gilsá event. They proposed that the Olduvai event should take a terminological precedent, with the event or events at about $t = 2.0$ my being redesignated as the Réunion event(s) after the first unambiguous detection [3, 13]. It is not yet clear that all these suggestions can be adopted, because it is far from certain that the Gilsá and Olduvai events are in fact one and the same. Nevertheless it is clearly desirable to define precisely the limits of the Réunion event.

3. Geological setting

Réunion is a wholly volcanic island built by eruptions from two large shield volcanoes, Piton des Neiges and Piton de la Fournaise, the geology of which has been summarized by Upton and Wadsworth [19] and McDougall [20]. The northwestern two thirds of Réunion is occupied by Piton des Neiges, an extinct volcano which has relatively well preserved external constructional slopes. The northwestern sector of Piton des Neiges, from St. Denis to La Possession and extending south, is an upstanding massif (Grande Montagne), consisting of a sequence of olivine rich basalt lavas of the Oceanite Series, which extend from sea level to an altitude of 1500 m (fig. 1). In general these lavas have shallow seaward dips, rarely exceeding 5°. Normally the individual flows are in the range 1 to 5 m thick.

Sixteen olivine basalts (14 lavas, 2 dykes) from the Grande Montagne massif yielded K–Ar ages of between 1.91 and 2.08 my [20], all but two lying in the even narrower age range of 1.95 to 2.06 my. These data indicate that eruption of the thick flow sequence occurred rapidly over a time interval probably not exceeding 0.1 my. Samples of 38 flows and dykes from the 2 my old Grande Montagne olivine basalts were measured palaeomagnetically by Chamalaun [21], who showed that 12 had normal and 26 reversed polarity. A polarity change was recorded in each of two lava sequences in the western wall of the 600 m deep canyon cut by Rivière St. Denis, adjacent to the eastern margin of the Grande Montagne massif. The detailed palaeomagnetic and geochronological study reported here was made on samples from these two sequences which are about 3 km apart, and which, for convenience, are termed the Rivière St. Denis sequence and the Grande Chaloupe sequence (fig. 1).

The accessible part of the Grande Chaloupe section consists of about 10 lava flows well exposed in a cut on the Chemin Arnoux (road C.F. 1) to Plaine d'Affouches, at an altitude of about 1130 m, high up on the valley wall overlooking Ilet à Guillaume (fig. 1). The lavas in general have a gentle northerly dip, the road traversing the sequence over a distance of about 700 m. The lavas are partly fragmental tabular flow units 2 to 5 m thick. Several of the flow units can be seen to pinch out laterally.

The second sequence sampled is exposed along the track into Rivière St. Denis from above St. Bernard. The section commences at an altitude of about 770 m on the western rim of the canyon, and sampling of the tabular flow units was carried out down to an altitude of about 500 m (fig. 1).

From the low dips of the lavas and the lack of obvious breaks in the valley wall between the two sequences it is clear that the Grande Chaloupe section is stratigraphically above the Rivière St. Denis section. L. Montaggioni (personal communication) recently reexamined the relevant part of the canyon of the Rivière St. Denis and confirms that our interpretation of the stratigraphic relationships is correct.

4. Methods

4.1. Grande Chaloupe sequence

One or two geographically oriented hand specimens were collected from 5 of the flows by McDougall in 1964. Palaeomagnetic measurements made by Chamalaun on these samples indicated a polarity change from normal (older) to reversed (younger), and because of this the locality was revisited in 1965 by McElhinny who collected 3 oriented samples from each of 7 flows. Chamalaun [21] included most of the results in his palaeomagnetic study of Réunion lavas, but mention of the polarity change was made only in passing. All the relevant data therefore are reported here (table 1). The techniques of palaeomagnetic measurements used were described previously [21] and included a demagnetization experiments to 150 Oe to minimize the effects of unstable components. A number of additional K–Ar isotopic age measurements have been made on these lavas for the present study.

4.2. Rivière St. Denis sequence

In 1965 McElhinny collected one oriented sample from each of 6 flows distributed throughout the top 30 flows exposed along the track into the canyon (fig. 1). Results from these samples (Chamalaun, personal communication) showed that a polarity change from reversed (older) to normal (younger) occurred in the section. An age of 2.03 ± 0.02 my was recorded [20] on the stratigraphically lowest sample collected (GA 1272). Subsequently in 1972 Watkins drilled

TABLE 1
Palaeomagnetic results

No.	B	D	I	R	α_{95}	θ'	ϕ'	Polarity
Grande Chaloupe Section								
GC1	4	200.7	27.5	3.971	9.1	− 69.2	310.3	R
3	3	176.1	16.6	2.954	18.9	− 76.9	217.7	R
4	3	193 3	34.4	2.894	29.3	− 77.3	318.3	R
7	3	180 9	31.0	2.981	12.2	− 85.7	247.5	R
8	4	Random		1.121	–	–	–	–
9	3	9.2	− 47.2	2.987	10.0	78.8	188.5	N
10	3	4.0	− 33.1	2.931	23.4	85.2	108.4	N
Rivière St. Denis Section								
RN2	3	354 4	51.4	2.905	27.5	36.8	49.5	T
3	3	356 0	32.7	2.805	40.6	51.1	49.3	N
4	3	5 9	− 24.0	2.995	6.4	79.9	90.3	N
5	3	2.8	− 8.5	2.997	5.0	73.2	65.2	N
6	3	13 3	− 31.4	2.984	11.2	76.8	130.4	N
7	3	8.5	− 38.9	2.995	6.3	82.1	154.9	N
8	3	7.9	− 37.4	2.983	11.5	82.6	146.8	N
9	3	352.5	− 5.0	2.983	11.5	70.2	32.8	N
10	3	34.6	14.0	2.986	10.3	45.9	109.6	N
11	3	329.6	− 60.7	2.942	21.3	57.0	279.4	N
12	3	7.2	− 55.2	2.941	21.5	73.9	213.7	N
13	3	356.9	− 38.8	2.987	9 8	87.0	304.9	N
14	3	181.6	30.8	2.992	7.9	− 85.4	254.9	R
15	3	186.6	48.1	2.850	35.2	− 79.8	20.6	R
16	3	178.1	31.8	2.995	5.9	− 85.9	209.3	R
17	3	182.6	53.3	2.993	7.4	− 76.9	46.0	R
18	3	209.1	56.4	2.986	10.5	− 60.1	4.3	R
19	3	167.7	49.9	3.000	1.2	− 75.2	101.4	R
20	3	164.8	53.5	2.999	3.0	− 71.2	97.8	R
21	3	190.8	76.6	2 998	3.6	− 45.7	48.8	R
22	3	175.8	46.8	2.992	8.0	− 81.9	82.8	R
23	3	161.0	39.0	2 993	7.5	− 72.3	138.2	R
24	3	182.6	50.5	2.972	14.7	− 79.4	43.1	R
25	3	160.3	60.0	2.958	18.2	− 63.9	90.9	R
26	3	198.3	25.5	2.986	10.5	− 71.0	305.2	R
27	3	205.2	19.1	2.846	35.6	− 63.4	304.6	R
28	3	187.7	41.8	2.989	9.3	− 82.2	351.3	R
29	3	169.9	46.1	2.996	5.8	− 78.7	108.0	R

No. = number of lava (number increases with increasing stratigraphic age); B = number of separate specimens; D and I = mean declination and inclination of remanent magnetism, in degrees east of north, and with respect to the horizontal, respectively. Negative inclination is upward. R = resultant vector, applying unit vector per specimen. α_{95} = 95% confidence circle. θ' and ϕ' are latitude and longitude of virtual geomagnetic pole: θ is negative for southern hemisphere; ϕ is in degrees east. Polarity is normal (N), reversed (R), or transitional (T) which we define as occurring when the VGP latitude is less than 45°.

Grande Chaloupe Section data from Chamalaun [21] and Chamalaun (personal communication). Sample numbers are: flow GC1 – RU25, 26, 149, 150; GC3 – RU152, 153, 154; GC4 – RU155, 156, 157; GC7 – RU23, 172, 173; GC8 – RU19, 163, 164, 165; GC9 – RU166, 167, 168; GC10 – RU169, 170, 171.

TABLE 2
Potassium-argon age data for whole rock basalt samples, Grande Montagne region, Réunion Island

Lab. no.	Field no.	Flow no.	K (wt. %)	Radiogenic ^{40}Ar (10^{-12} mol/g)	100 Rad.^{40}Ar / Total ^{40}Ar	Calculated age (my) ± 1 sd	Polarity	Arithmetic mean age of flow (my) ± 1 sd	Weighted mean age (my) ± 1 se
Grande Chaloupe Section									
GA1273A	RU23	GC7	0.578, 0.578	2.062	29.6	2.00±0.02	R	–	2.02±0.02
GA2076	RU165	GC8	0.645, 0.650	2.296	51.5	1.99±0.02			
				2.284	47.7	1.98±0.03	–	2.00±0.03	
				2.347	53.6	2.03±0.02			
GA2077	RU170	GC10	0.578, 0.580	2.092	43.1	2.03±0.02	N	2.04±0.02	
				2.120	40.7	2.05±0.02			
Rivière St. Denis Section									
72-744		RN3	0.658, 0.661	2.370	35.1	2.02±0.02	N	2.01±0.01	2.00±0.01
				2.343	41.4	2.00±0.02			
				2.364	37.9	2.01±0.02			
72-775		RN4	0.463, 0.460	1.637	38.1	1.99±0.03	N	1.99±0.06	
				1.592	43.4	1.94±0.03			
				1.681	41.0	2.05±0.02			
72-484		RN12	0.646, 0.648	2.286	54.7	1.98±0.02	N	–	
72-485		RN13	0.631, 0.631	2.192	9.1	1.95±0.08	N	–	
72-486		RN14	0.447, 0.447	1.576	29.4	1.98±0.02	R	–	2.02±0.02
72-487		RN15	0.711, 0.711	2.527	33.1	2.00±0.03	R	2.05±0.05	
				2.650	37.5	2.09±0.02			
				2.620	35.9	2.07±0.02			
72-488		RN28	0.412, 0.413	1.519	40.0	2.07±0.02	R	–	
72-489		RN29	0.789, 0.791	2.937	43.1	2.09±0.02	R	2.04±0.07	
				2.807	43.2	2.00±0.02			

three separate cores of average length 12 cm and 2.5 cm in diameter from the top 29 flows in the same section. Each core was oriented in geographic coordinates to a precision of ± 3°. Specimens 2.2 cm in length were used for the palaeomagnetic measurements. Unstable components were minimized using af fields up to 300 Oe [22], and directions and intensities of remanent magnetization were measured using a spinner magnetometer. Conventional Fisher statistical methods [23] were used to analyse these data, listed in table 1.

4.3. Potassium-argon dating

Petrographic examination was made of thin sections from all the available samples in order to determine those most suitable for K–Ar dating. All the samples were olivine basalts with phenocrysts or microphenocrysts of olivine, clinopyroxene, plagioclase and rare iron oxide, set in a groundmass of the same minerals. Except for several samples which showed extensive alteration of the groundmass the rocks were virtually free of alteration and were mainly holocrystalline. Only rocks showing no or little alteration were used for the age determinations to minimize the possibility of diffusional loss of radiogenic argon. Several samples showed minor (< 5%) intersertal mesostasis, or more rarely, fresh isotropic glass. Techniques, of K–Ar age measurement were similar to those described previously [20, 24]. Potassium was determined by flame photometry, and argon was measured by isotope dilution on samples of 15 to 35 g. The ^{38}Ar tracer used was from a gas pipette system; over 40 calibrations of this tracer indicate a standard deviation of ± 0.48% for a single tracer calibration. The isotopic composition of the extracted argon was measured either in a Reynolds-type mass spectrometer, or in a

substantially modified AEI MS10 mass spectrometer which employed a 4.2 kG permanent magnet and digital readout. The K–Ar ages are given in table 2 and are calculated using the following decay constants – $\lambda_e = 0.585 \times 10^{-10}$ yr^{-1}, $\lambda_\beta = 4.72 \times 10^{-10}$yr^{-1}, $^{40}K/K = 1.19 \times 10^{-2}$ atom %. The precision of each age measurement is given at the level of one standard deviation based upon the statistically derived uncertainties for each argon and potassium measurement as described by McDougall et al. [24]. These do not take into account uncertainties in the decay constants. The standard deviation of precision averages 1.25%, including one value of 3.0% from a sample (72-485) in which the proportion of radiogenic argon was less than 10%. An alternative way of estimating precision is by pooling data from those samples for which replicate measurements were made. This yields an average value of 1.13%, in very satisfactory agreement with the value determined from the individual age measurements. Where more than one age determination was made on a particular sample a mean age and standard deviation has been calculated (table 2). In calculating a mean age for a number of lavas the data have been combined by a weighting procedure according to the inverse of the variances to allow for the differing precision of the ages, yielding a weighted mean age and standard error. The generally excellent agreement of the ages suggests that diffusional loss of radiogenic argon is minimal or absent.

5. Results

5.1. Grande Chaloupe section

The stratigraphically highest four flows sampled in the Grande Chaloupe section have reversed polarity, whereas the lowest two flows have normal polarity (table 2). The high latitude of the virtual geomagnetic poles (VGPs) enables confident identification of a true change in polarity of the geomagnetic field. Flow GC8, which lies stratigraphically between the normal and reversed polarity lavas, gives scattered directions of magnetization on four samples (table 1). After af demagnetization to 150 Oe these samples had an average intensity of 1.1×10^{-4} emu cm^{-3}, a factor of 4 to 15 times less than the intensity of the comparably treated normal and reversed polarity flows in the Grande Chaloupe sequence (Chamalaun, personal communication). The samples from GC8 are fresh and petrographically similar to the associated lavas. All these data make a very strong case for lava flow GC8 having been erupted during the polarity transition. It is of interest to note that the upper surfaces of flows GC8 and GC9 are conspicuously reddened by oxidation, suggesting a rather longer interval between eruptions than was the case in the remainder of the section.

Potassium–argon ages from three lavas of the Grande Chaloupe sequence are given in table 2. These age measurements were made for the present study, and supersede those given earlier [20] on the same samples measured in 1966. The two sets of calculated ages on the three samples differ by from 1 to 2.5%. Re-measurement was undertaken so that direct and meaningful comparisons could be made with the ages on the samples from the Rivière St. Denis section, which were measured at the same time. Such an approach overcomes any possible systematic difference or error in the age measurements between 1966 and 1973.

Ages on samples from the reversed polarity lava flow GC7 and the randomly magnetized flow GC8 are indistinguishable at 2.00 ± 0.02 my. The normal polarity flow GC10 yields an age of 2.04 ± 0.02 my which is not significantly different from the ages of GC1 and GC8 at the 95% level of confidence [25]. These data are interpreted to indicate that the polarity change occurred over an interval of time less than 30,000 yr. This figure is an estimate based upon twice the standard error of the weighted mean age of the three flows dated. While statistically not significant the apparent difference in age between the reverse and normal polarity flows is also of this same order (40,000 yr). Our estimate of the actual age of the polarity change is 2.02 ± 0.02 my.

5.2. Rivière St. Denis section

Palaeomagnetic data from this sequence (table 1) show that a polarity change from reversed (older) to normal (younger) occurred between the extrusion of lava numbers 13 and 14. The high latitude of the VGPs shows that we are dealing with a true reversal of the polarity of the geomagnetic field. The stratigraphically youngest lava sampled (RN 2) in this sequence exhibits transitional polarity (table 1, fig. 2); it probably was

erupted during the time when the polarity was changing from normal to reversed as recognized in the Grande Chaloupe section.

Ages were measured on samples from 8 of the 29 flows collected (table 2), two on lavas from near the top of the section, two on flows from near the bottom and four on lavas adjacent to the level of the polarity change. The four normally magnetized lavas that were dated give ages that are indistinguishable from one another at the 95% confidence level. The weighted mean age and standard error when the results from the four lavas are combined is 2.00 ± 0.01 my. Three of the four reversed polarity lavas dated yielded calculated ages that agree to within experimental error, but the measured age on RN 14 is significantly younger. With such good agreement between three of the four lavas it is considered justifiable to combine all four ages, which yields a weighted mean age and standard error of 2.02 ± 0.02 my (table 2).

It is clear that there is no significant difference in weighted mean age between the reversed polarity lavas and the overlying normal polarity lavas. From these data we estimate the age of the polarity change to be 2.01 ± 0.01 my, which is indistinguishable from the age of 2.02 ± 0.02 my arrived at for the polarity change of opposite sign in the Grande Chaloupe sequence.

It is apparent from the age data that the time involved in the eruption of the 29 lavas of the Rivière St. Denis section is relatively short, possibly not much greater than the uncertainties of some of the ages. Because of the limited resolution of the dating quantitative estimates of duration of the volcanism are difficult to make. Perhaps a reasonable upper limit of the duration is given by the difference between the maximum and minimum measured ages in the sequence, 120,000 yr. Inspection of the data in table 2 and the lack of difference in age between the weighted average ages for the reversed and normal polarity lavas would indicate that a period of less than 60,000 yr for the eruption of the complete sequence of lavas sampled may not be unreasonable. Thus an average time interval of 2000 yr or less between successive flows could be calculated; this is consistent with an interval of at least 1000 yr between eruption of lavas RN 13 and RN 14 from the fact that no transitional directions are found in lavas adjacent to the polarity change [26].

6. Discussion

As previously discussed we are confident that the Grande Chaloupe sequence stratigraphically overlies the Rivière St. Denis sequence in the west wall of the canyon (fig. 2). The stratigraphy together with the palaeomagnetism and geochronology clearly indicate that a normal polarity interval has been recorded in the lavas (fig. 2). This normal polarity interval is here defined as the Réunion normal event. From the general concordance of the K–Ar ages this event may be as short as 10,000 yr or as long as approximately 50,000 yr. The relatively short period during which the geomagnetic field had normal polarity no doubt is the reason why this event has been recorded only rarely in palaeomagnetic studies in deep sea sedimentary cores [6].

In our previous studies of Réunion lavas [3, 13, 20] several other normal polarity samples with ages of about 2 my were found in the Grande Montagne region (GA 1268, 2.01 ± 0.04 my; GA 1289, 1.97 ± 0.03 my, a different sample from the same flow as GA 1268; GA 2072, 1.98 ± 0.08 my; GA 1269, 1.96 ± 0.03 my). Lack of outcrop and possible faulting prevents direct correlation in the field of these samples with the lavas of the Réunion event. No significant difference is found, however, between the K–Ar ages of these samples and the mean age of the Réunion event. Thus we have no direct evidence in Réunion of more than one polarity event of about this age.

Normal polarity lavas of about the age of the Réunion event have been recorded from several other regions including St. George Island, Alaska (ages of 1.95 ± 0.10 my and 1.96 ± 0.04 my) [11], Cerro del Fraile, Argentina (ages of 2.05 ± 0.02 my and 2.06 ± 0.03 my) [27], and Cocos Island (an age of 2.09 ± 0.04 my) [12]. It is possible that all these normal polarity volcanics were erupted during the Réunion event as designated here, because all the K–Ar ages have errors that overlap or nearly overlap the age of the event. Our results do not confirm or deny the suggestion [1, 18] that there are two normal polarity events close to $t = 2$ my.

The age of the Réunion event as defined here corresponds closely to the estimated age of 1.99 ± 0.06 my or 1.97 my for the *W* anomaly identified by Emilia and

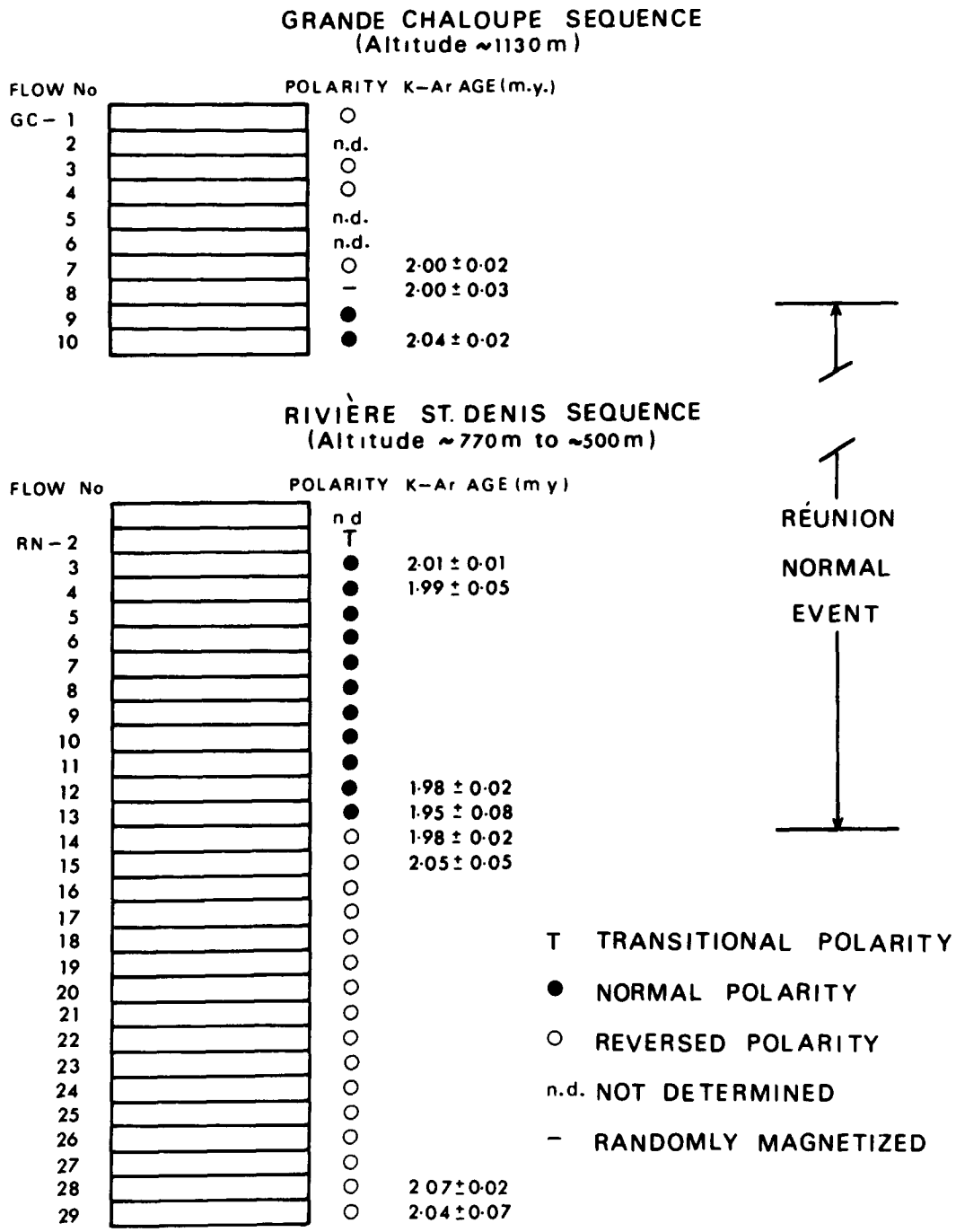


Fig. 2. Diagram that schematically shows the lavas of the two sections, their polarity and their measured ages.

Heinrichs [28, 29] in magnetic anomaly profiles across mid ocean ridges. Another anomaly, X, recognized by Heirtzler et al. [30] has an estimated age of 2.3 ± 0.1 my [28], which is significantly older than the Réunion event.

In summary, the present results from the Grande Montagne area of Réunion indicate that the Réunion normal polarity event within the Matuyama reversed epoch has a mean age of 2.02 ± 0.02 my, and a duration within the range 10,000 yr to 50,000 yr. The Réunion event clearly is older than the Olduvai normal event [18], and of much shorter duration.

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